

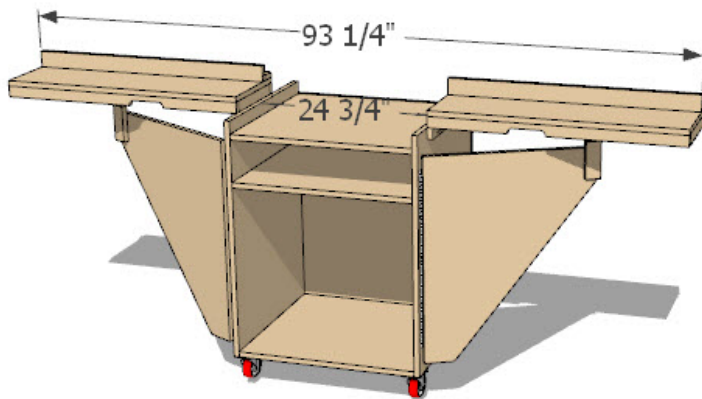


FIX THIS BUILD THAT

Mobile Miter Saw Station

The Mobile Miter Saw Station is made from 2 sheets of $\frac{3}{4}$ " plywood. It is 39" tall on 3" casters. With side wings extended it's 93-1/4" wide and will support just under 4' of stock on each side, allowing you to make end cuts on 8' stock with full support.

A basic version built out of common plywood can be made for right around \$100. But upgrades such as better plywood, larger casters, dust collection, an automated vacuum switch, and an aftermarket fence and stop system make this build a workshop masterpiece.

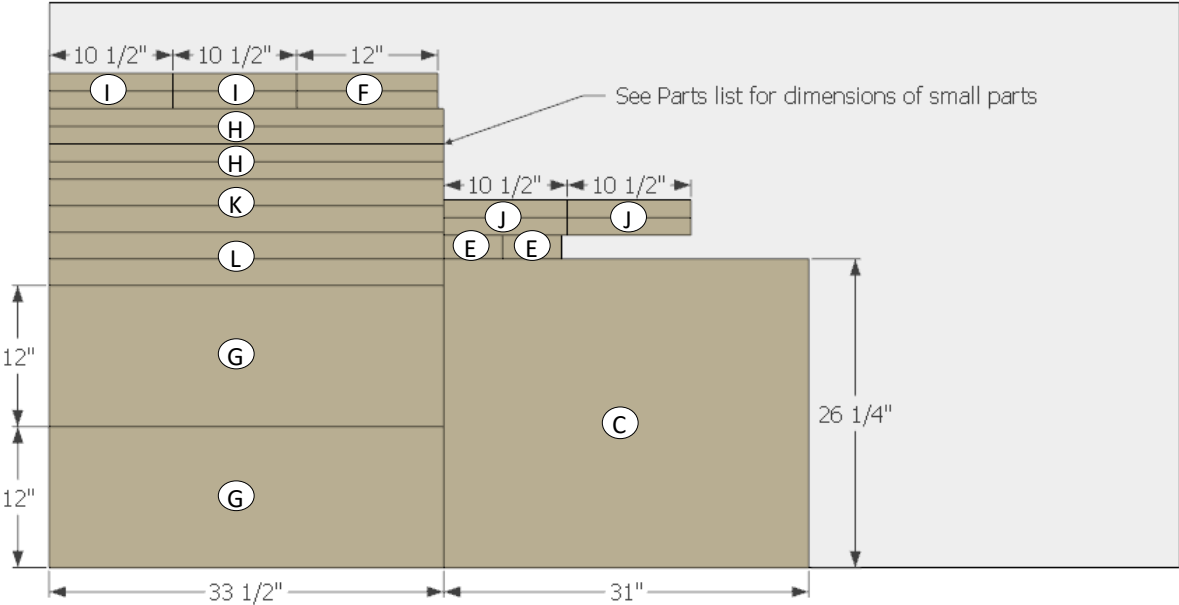
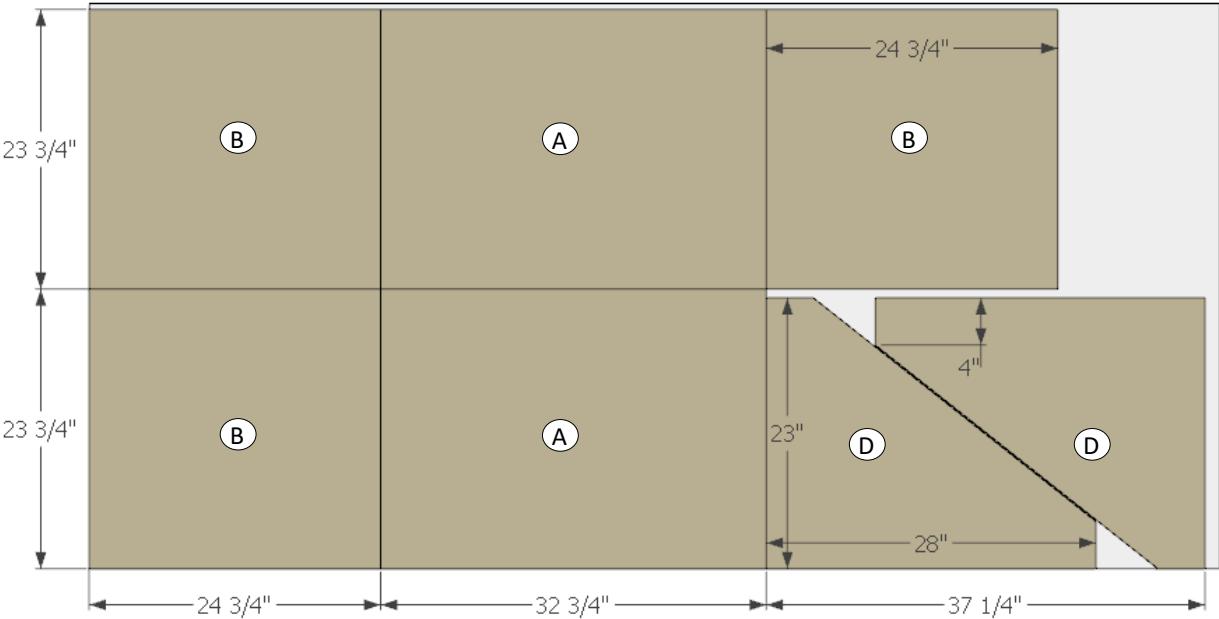


fixthisbuildthat.com/mitersawstand

Click Here for Build Video

Cutting Diagram and Parts List

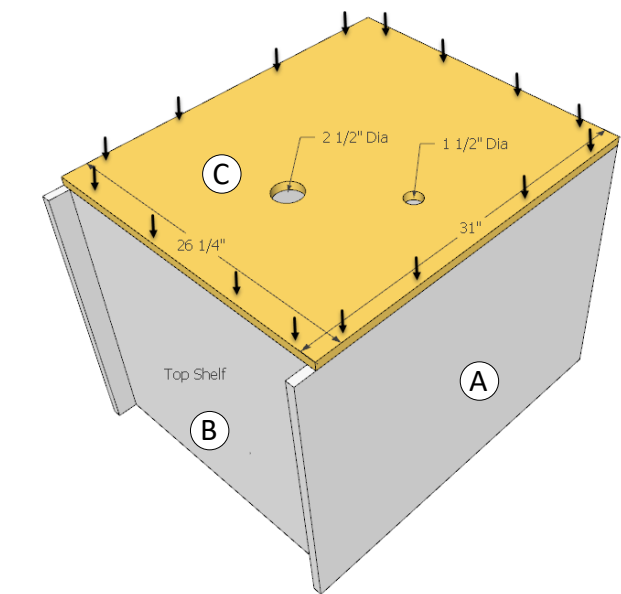
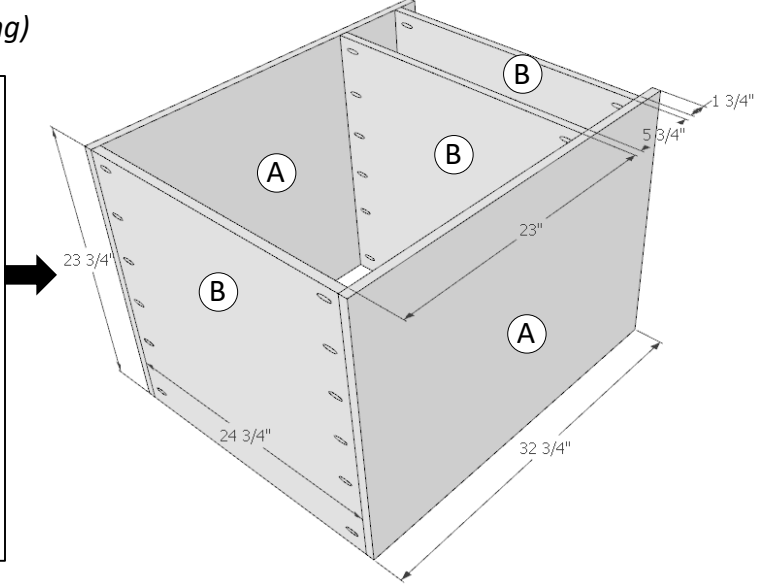
Part	Description	Qty	Dimensions	Material
A	Side	2	32-3/4" x 23-3/4"	3/4" Plywood
B	Shelf	3	23-3/4" x 24-3/4"	3/4" Plywood
C	Back	1	31" x 26-1/4"	3/4" Plywood
D	Side Wing Support	2	28" x 23" (triangular)	3/4" Plywood
E	Support Riser	2	5" x 2"	3/4" Plywood
F	Side Wing Mounting Cleat	2	12" x 1-1/2"	3/4" Plywood
G	Side Wing Top	2	33-1/2" x 12"	3/4" Plywood
H	Side Wing Long Frame	4	33-1/2" x 1-1/2"	3/4" Plywood
I	Side Wing Short Frame	4	10-1/2" x 1-1/2"	3/4" Plywood
J	Support Blocks	4	10-1/2" x 1-1/2"	3/4" Plywood
K	Fence Back Rail	2	33-1/2" x 2-1/4"	3/4" Plywood
L	Fence Front Rail	2	33-1/2" x 2-1/4"	3/4" Plywood
M	Hardwood Spacer	2	As needed for your saw	Hardwood



Build Instructions: *(Read all steps before starting)*

Step 1: Cut 2 sides (A) and 3 shelves (B) to size per cutting diagram. Drill pocket holes for 1-1/4" pocket hole screws on the underside of each shelf. Start with the top shelf and attach it to both sides 1-3/4" down from the top. Attach the middle shelf 5-3/4" down from the top shelf and the bottom shelf is flush with the bottom of the sides.

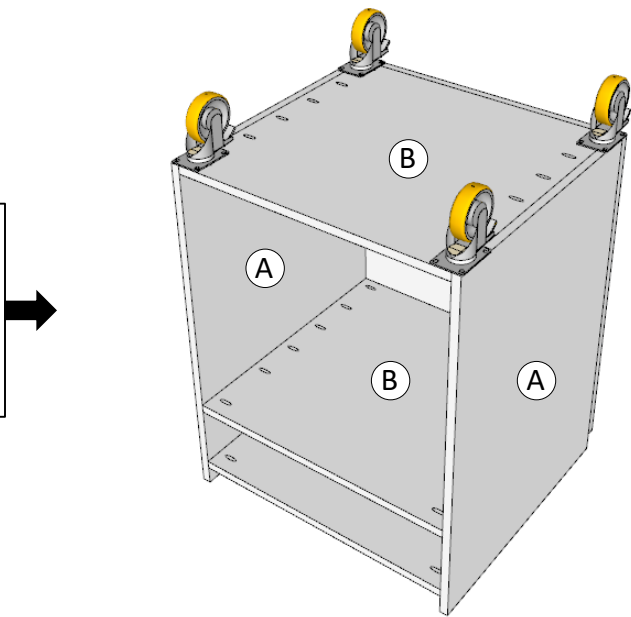
NOTE: *Make sure your miter saw base dimensions will fit in this footprint. If your saw is wider than 24-3/4" or taller than 4" then you may need to adjust the dimensions of parts A, B and C*



Step 2: Cut a back (C) to size. Lay the base flat on it's front and square up the assembly. Secure the back (C) to the base with 1-1/4" wood screws, predrilling and countersinking first. The back will overlap the top shelf and bottom shelf but not go to the top of the base.

If you will be using dust collection, drill a 2-1/2" hole in the back (C) for the vacuum hose and a 1-1/2" hole for the power cord(s). Holes should be below middle shelf. Round all edges with a router and roundover bit.

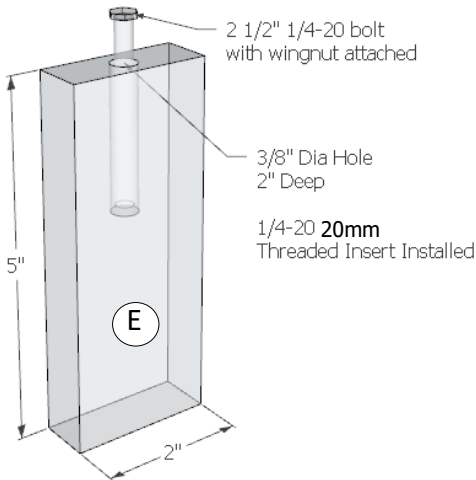
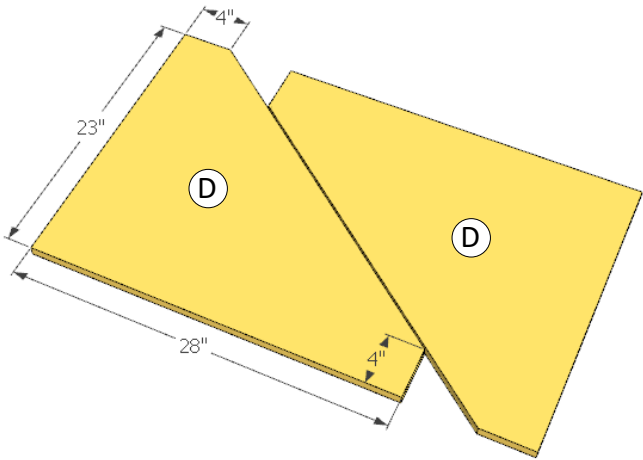
Step 3: Flip the base assembly upside down on its top and attach the casters in the corners with screws and 1/4" washers. Use 1-1/4" screws when screwing into the sides and back. Use 5/8" screws when going into the bottom shelf.



Build Instructions: *(Read all steps before starting)*

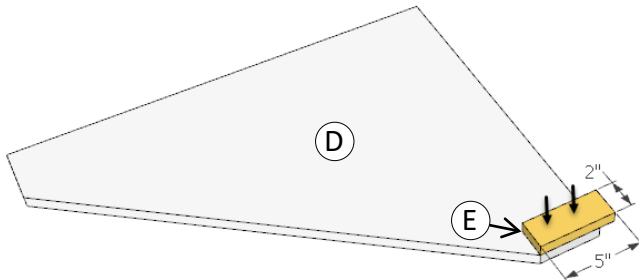
Step 4: Cut a piece of ply 37-1/4" x 23". Using a circular saw and straight edge, cut diagonally across the piece (offset by 4" as shown to the left), to make 2 Side Wing Supports (D).

Cut the triangular corners off to get 28" tall pieces. Round edges with a router and roundover bit.

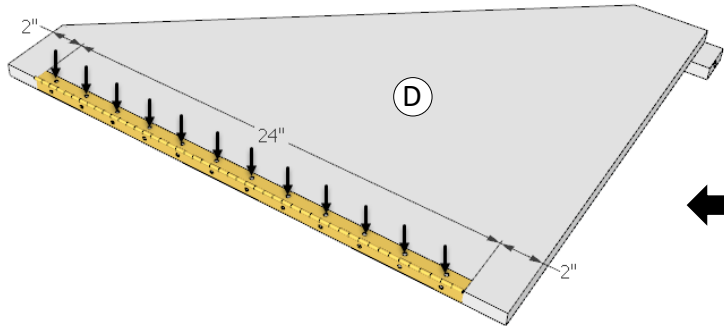


Step 5: Cut 2 Support Risers (E) to size. Drill a 3/8" hole 2" deep in the end of each riser. Install a 1/4"-20 x 20mm threaded insert into each riser using a hex wrench. Thread in a 2-1/2" hex bolt with wing nut into each insert. Wing nut is used to lock bolt in place and optional lock washer can be added under wing nut.

Step 6: Attach the Support Risers (E) to the Side Wing Supports (D) with glue and 1-1/4" screws. The Risers should be flush with the short edge and stick up 1" past the top of the Side Wing Support. Note, left side will be a mirror image of the right side with the riser on the front of the wing.



(right side wing support shown)



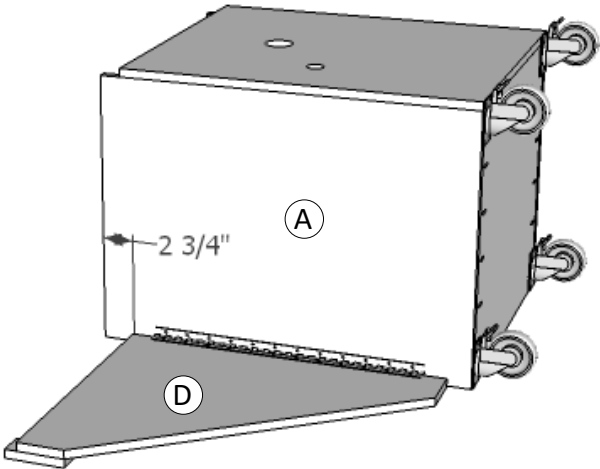
(right side wing support shown)

Step 7: Cut two 24" pieces from a 72" long 1-1/2" piano hinge. Center and install the hinges on the long edge of the Side Wing Supports (D). Lay the hinges over the corner of the edge to keep it square. Hold firmly and secure with 5/8" screws.

Build Instructions: *(Read all steps before starting)*

Step 8: Join the Side Wing Supports (D) to the side (A) using 5/8" screws through the piano hinges. The Supports should be 2-3/4" down from the top of the base and flush with the front of the base when open at 90 degrees.

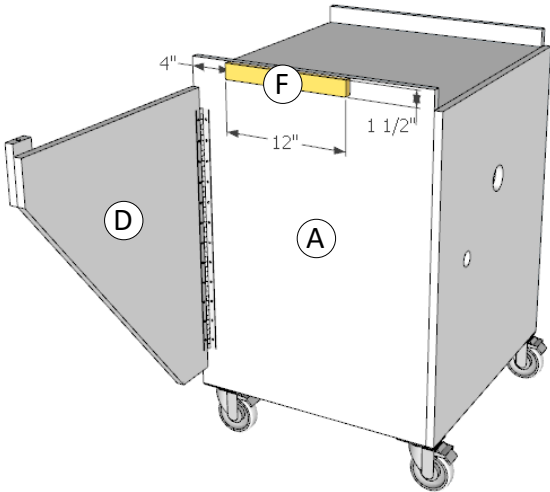
Laying the base on it's front with offcut plywood spacers under it will make installation easiest. Watch the video on FixThisBuildThat.com for a visual on how to do this.



(right side wing support shown)

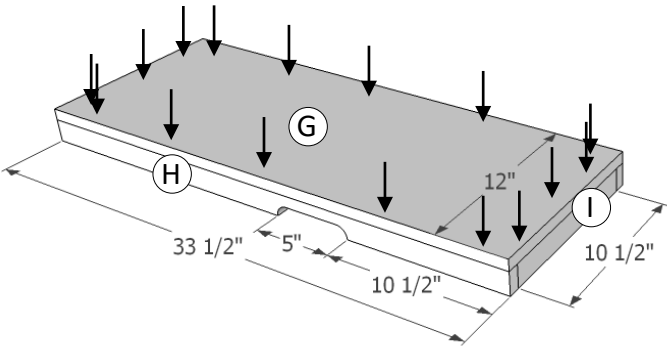


Step 9: Cut 2 Side Wing Mounting Cleats (F) to size and attach to the sides of the base with 1-1/4" screws. Cleats should be flush with the top and 4" back from the front.



Step 10: Cut 2 Side Wing Tops (G), and 4 each of the Side Wing Long Frame (H) and Short Frame (I) pieces to build 2 side wings. Secure the Top to the Long and Short frame pieces using 1-1/4" screws to make the side wing assembly. Round edges with a router and roundover bit.

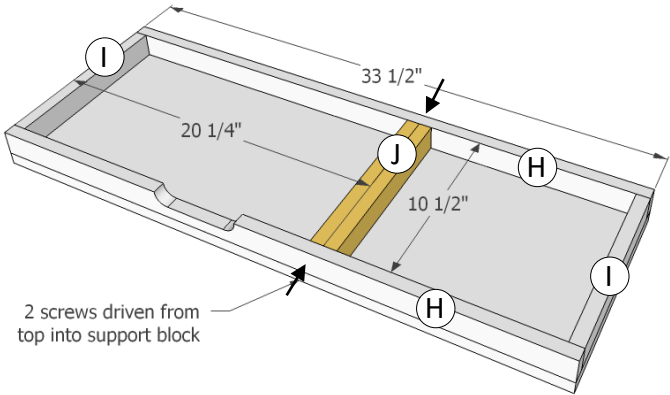
Add a small hand hold to the front side of each wing in one of the Long Frame (H) pieces. Cut out is 3/4" tall and can be made before or after assembly.



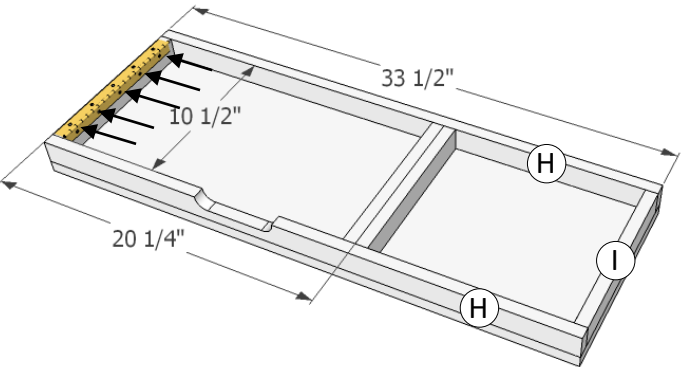
(left side wing shown)

Build Instructions: *(Read all steps before starting)*

Step 11: Cut 4 Support Blocks (J). Glue and/or screw 2 blocks together face to face to make 2 pair. Turn over the Side Wing and install a pair of Support Blocks (J) between the Long Frames (H) with screws from each side and 2 from the top. The center of the Support Blocks should be 20-1/4" from the outside end of the Side Wing.



(left side wing shown)



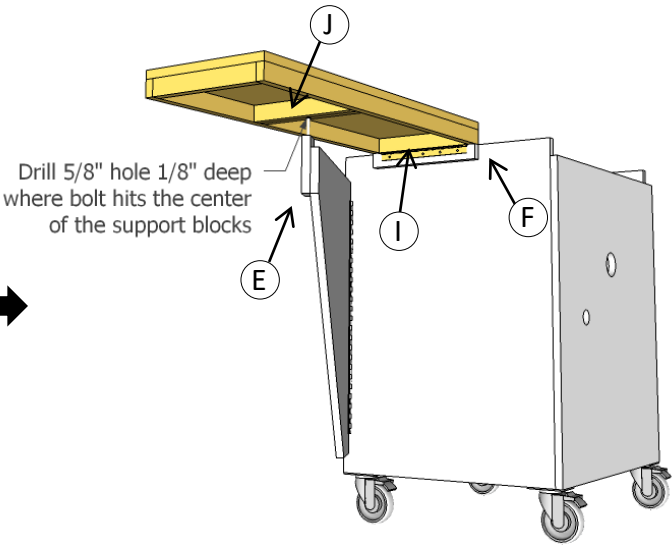
(left side wing shown)

Step 12: Cut two 10-1/2" sections from the piano hinge. Center and install the hinges on the Side Wing Short Frame (I) pieces with 5/8" screws attached to the inside face of the Short Frame (I).

Step 13: Center the Side Wing assembly on the Side Wing Mounting Cleat (F) with the Short Frame (I) positioned directly on top of the Cleat (F). Clamp the assembly in place. Rest the Support Blocks (J) on the leveling bolt of the Support Riser (E). Before attaching the piano hinge, remove the Side Wing Assembly and drill a 5/8" hole 1/8" deep where the bolt hits the center of the support blocks (J). This will index the bolt.

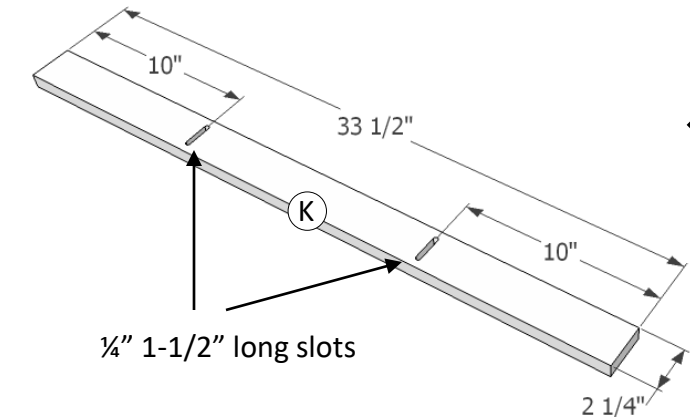
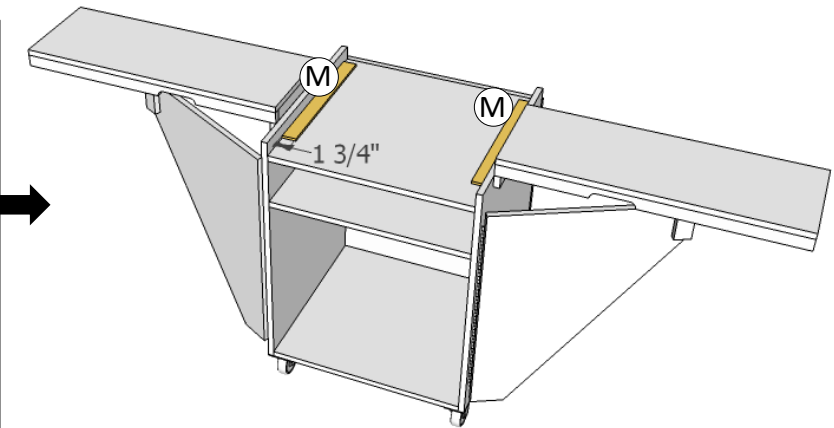
Put the Side Wing assembly back on the Mounting Cleat (F) and attach the piano hinge to the Cleat using 5/8" screws.

Watch the video on FixThisBuildThat.com for a visual.



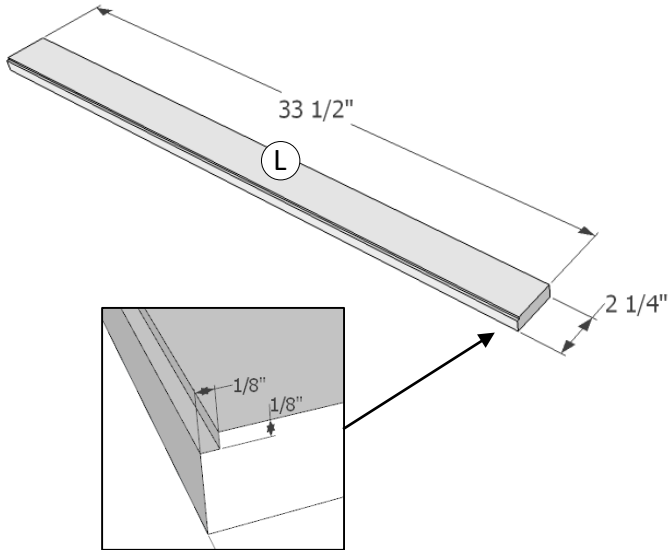
Build Instructions: *(Read all steps before starting)*

Step 14: Cut 2 hardwood spacers (M) to size based on your miter saw. Level the Side Wings and put the saw on top of the stand. With a level extended across both Side Wings, measure the distance between the bottom of the level and the miter saw deck. This is the thickness to cut the spacer (M). The length and width should be determined by the spacing of the saw mounting holes. Put the spacers under the saw and secure the saw with screws or bolts, washers, and threaded knobs.



Step 15: Cut 2 Fence Back Rails (K). For an adjustable fence, route 1/4" wide by 1-1/2" long slots in the Back Rails (K) 10" from each end.

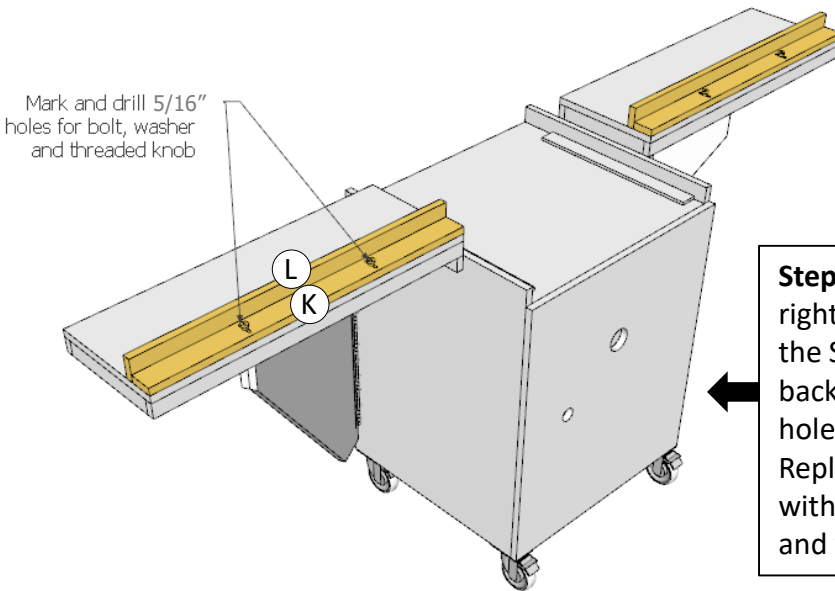
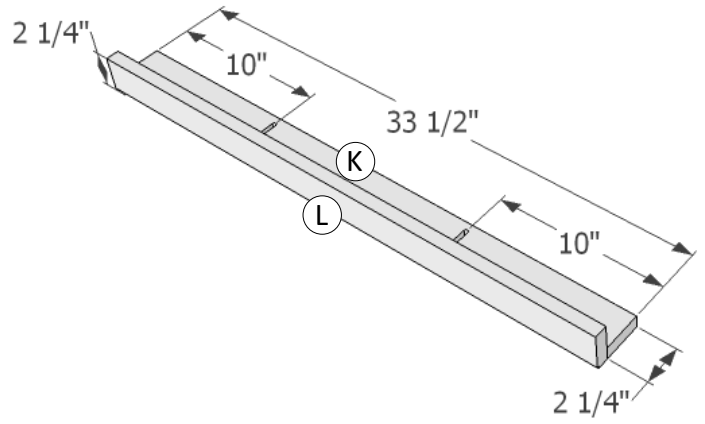
Step 16: Cut 2 Fence Front Rails (L). For optional dust relief on the fence cut a 1/8" rabbet on the front lower edge of the Front Rails (L) as seen in inset. This will prevent sawdust from getting between the fence and workpiece.



Build Instructions: *(Read all steps before starting)*

Step 17: Attach Front Rail (L) to Back Rail (K) using 1-1/4" screws, predrilling and countersinking first.

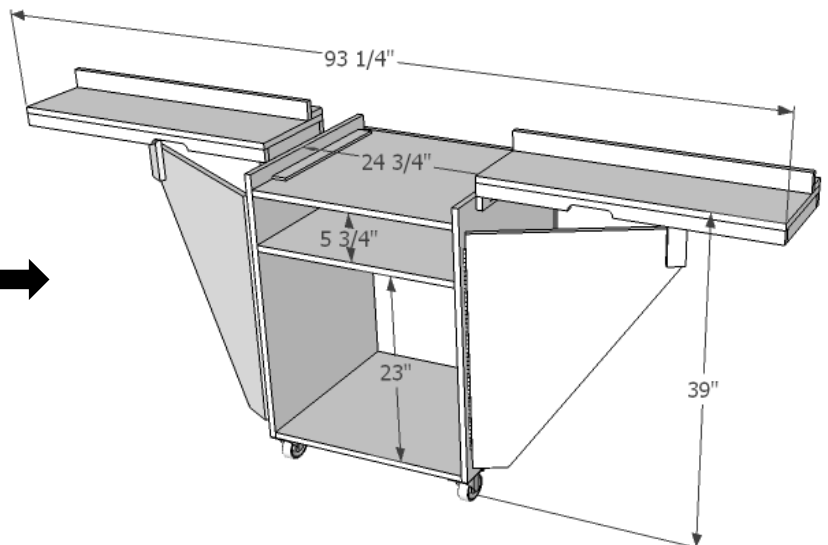
Make sure fence is at 90 degrees before securing.



Step 18: Use a 4' straight edge to align the left and right fences with the miter saw fence. Mark a spot on the Side Wings through the center of each slot on the back fence rail (K). Remove the fences and drill 5/16" holes through the Side Wings at the marked spots. Replace and realign fences then attach with 2-1/2" bolts with lock washer and fender washer on the underside and washer and threaded knob on the topline.

Step 19: Add upgrades as desired such as an aftermarket fence stop system with stops, vacuum for dust collection, and dust collection automation.

Go to FixThisBuildThat.com to see examples of upgrades and other great projects.



[Click here for direct links to these items](#)

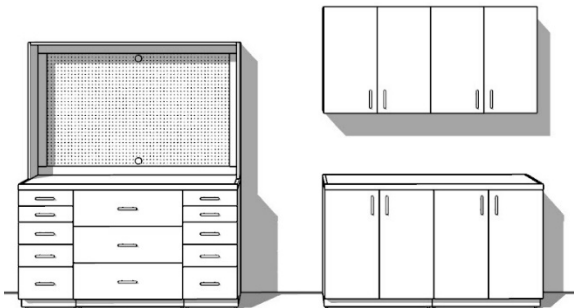
Tools Used

- 12" Dual Bevel Miter Saw
- 18v 5pc Toolkit
- Compact Router
- Brad Nailer
- 12 Gal Shop Vacuum (or 4 gallon and cyclone separator)
- Pro-grade Hose Kit

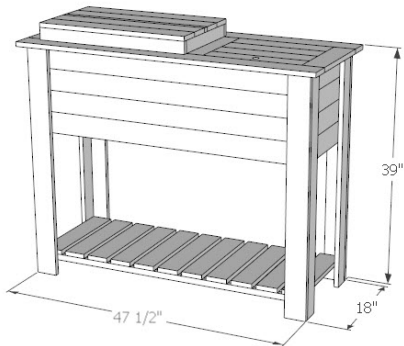
Materials/Supplies

- (2) 4'x8' 3/4" Plywood
- 1-1/4" pocket hole screws
- 1-1/4" wood screws
- 5/8" wood screws
- 72" long 1-1/2" piano hinge
- (2) 1/4"-20 20mm threaded inserts
- (10) 2-1/2" 1/4"-20 bolts
- (2) 1/4" wingnuts
- (8) fender washers
- (8) 1/4" flat washers
- (8) 1/4" lock washers
- (8) 1/4"-20 threaded knobs
- (4) 3" locking casters

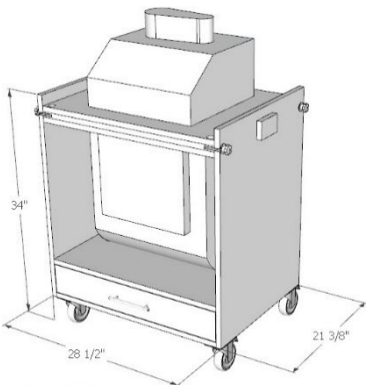
[Click Here for More of My Furniture and Shop Plans](#)



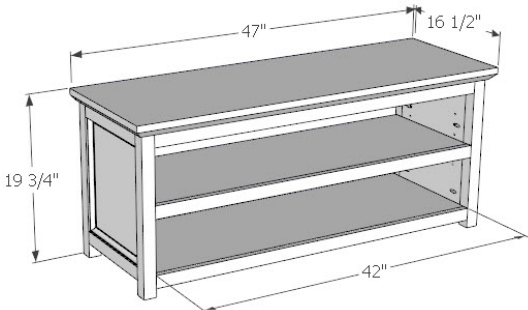
[Modular Garage Cabinets](#)



[Patio Cooler / Grill Cart](#)



[Flip Top Tool Stand for two tools](#)



[Shoe Storage Bench](#)